

Week	Task ID	Task Description	Subtasks
1	A1	Image Dataset Acquisition	Scrape/find 1,000+ labeled cat faces (Happy, Sad, Angry, Fear).
	A2	Image Preprocessing Pipeline	Script for resizing, grayscale, and normalization.
	A3	Augmentation Strategy	Implement rotation, flipping, and brightness shifts.
	A4	EDA: Image Class Balance	Visualize distribution of emotions; address imbalances.
2	A5	CNN Architecture Design	Set up ResNet or MobileNet base layers.
	A6	Model Training (Vision)	Train facial classification model.
	A7	Hyperparameter Tuning	Optimize learning rates and dropout.
	A8	Vision Model Evaluation	Generate Confusion Matrix and F1-scores.
3	A9	Backend API Development	Build Flask/FastAPI endpoints for image processing.
	A10	Integration: Image to Model	Connect API to the saved Vision model weight file.
	A11	Prediction Logic	Script to return confidence scores and labels.
	A12	Error Handling (Vision)	Handle "No Cat Detected" or low-quality uploads.
4	A13	System Integration Testing	Test end-to-end flow from upload to prediction.
	A14	Vision Performance Report	Document model strengths and edge cases.
	A15	Technical Documentation	Write README for model deployment.